Form 55 25 March 2027 Critical reading of media formulations

1. Warming · Flash [PRO] A 10% increase out of 40: what is the final value?

Rep. $40 \times 1{,}10 = 44$.

2. Heart · Critical Spirit [LEC] One article states: "60 per cent of accidents involve sober drivers". Can we conclude that driving sober is dangerous?

Rep. No: the reference population is missing. If the vast majority of drivers are sober, it is If they are the majority among the accident victims, we should compare conditional frequencies.

3. Heart + · True/false [LEC] True or false? Justify. "Two out of three smokers are men" means Two out of three men smoke.

Rep. Wrong. $_{\text{smoker}}(\text{man}\quad)$ and $_{\text{man}}(\text{smoker}\ )$: Reference populations differ.

4. Return · True/false [STA] True or false? Justify. The median of a series is always equal to its average.

Rep. False. Counter-example: 1, 2, 9: mean 4, median 2.

5. Red thread · Code reading [ALG] What is the value of `s` At the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```

Rep. $s = 1 + 2 + 3 + 4 = 10$.

Trap: range(1, 5) stop at 4 ; 5 is excluded.

Geometry identified: synthesis vectors / straight

Form 56 Mar. 30 March 2027 Vectors and colinearity: reinvestment

1. Warming · Flash [GEO] $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.

Rep. $\begin{pmatrix}2\\3\end{pmatrix}$; $\begin{pmatrix}6\\-4\end{pmatrix}$.

2. Heart · Method [GEO] Points $A(-1;2)$, $B(1;3)$ and $C(5;5)$ Are they aligned?

Rep. $\overrightarrow{AB}\begin{pmatrix}2\\1\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}6\\3\end{pmatrix}$: $2 \times 3 - 1 \times 6 = 0$: aligned.

3. Heart + · Method [GEO] $A(-1;2)$, $B(3;1)$, $C(4;4)$. Determine D such that $ABCD$ or a parallelogram.

Rep. $\overrightarrow{AB} = \overrightarrow{DC}$: $D = A + \overrightarrow{BC} = (-1+1;2+3) = (0;5)$.

4. Back · Flash [PRO] In a class of 24 students, 18 are half boarders. percentage?

Rep. $\dfrac{18}{24} = 0{,}75 = 75\,\%$.

5. Red thread · Prove [LIT] Show that $(n+1)^2 - (n-1)^2 = 4n$ for the integer n. What can we do about it? conclude for the difference of squares of two consecutive odds (n pair)?

Rep. $(n^2+2n+1) - (n^2-2n+1) = 4n$. For $n = 2k$ pair, the difference of two squares of odd consecutive is equal to $8k$: multiple of 8.

Both integers n -1 *et* $n+1$ are odd when n is even.

Form 57 game. 1st Apr. 2027 Determinant and Right Equations: Reinvestment

1. Warming · Flash [GEO] Slope on the right 3 $x + 2y - 6 = 0$?

Rep. $y = -\dfrac{3}{2}x + 3$: slope $-\dfrac{3}{2}$.

2. Heart · Method [GEO] Equation of the right (AB) with $A(1;2)$ and $B(4;8)$.

Rep. Pent $\dfrac{8-2}{4-1} = 2$; $y = 2x + p$, $2 = 2 + p$: $y = 2x$; or $2x - y = 0$.

3. Heart + · Method [GEO] Rights 2 $x - y + 1 = 0$ and $4x - 2y + 5 = 0$ are they parallel, combined Or stinging?

Rep. Director Vectors $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$ colinear: parallel; $(0;1)$ check the first but 4 $\times 0 - 2 + 5 = 3 \neq 0$: strictly parallel.

4. Back · Flash [Sighs] Calculate $10^{-2} \times 5 \times 10^{4}$.

Rep. $5 \times 10^{2} = 500$.

5. Red thread · Method [EQ] Solve in $\mathbb{R}$: $7 - 2(x-1) = 3x - 6$.

Rep. $x = 3$

Trap: $-2(x-1) = -2x - 2$ (sign error).

Form 58 Mon. 5 Apr. 2027 Flat geometry problems: choose a method

1. Warming · Flash [GEO] Distance AB for $A(0;0)$ and $B(5;12)$?

Rep. $\sqrt{25+144} = 13$.

2. Heart · Method [DEM] $A(0;0)$, $B(4;2)$, $C(-2;4)$. Nature of the triangle ABC ?

Rep. $AB^2 = 20$, $AC^2 = 20$, $BC^2 = 36 + 4 = 40 = AB^2 + AC^2$: isocele rectangle in A.

3. Heart + · Method [DEM] Identify two methods to demonstrate that $ABCD$ is a parallelogram to coordinates.

Rep. (i) $\overrightarrow{AB} = \overrightarrow{DC}$; (ii) diagonals $[AC]$ and $[BD]$ have the same background.

4. Back · Reading [LEC] In a class of 30 students, a stick diagram indicates x students for the footnotes 12 and $2x$ students for note 14; the other notes include 18 students. x.

Rep. $x + 2x + 18 = 30$, that's $3\,x = 12$ and $x = 4$.

5. Red thread · Flash [SOL] A pyramid has as its base a square side 6 cm and for height 5 cm. its volume.

Rep. $V = \dfrac{1}{3} \times 6^2 \times 5 = 60\ \text{cm}^3$.

Form 59 Mar. 6 Apr. 2027 Situations from other disciplines

1. Warming · Flash [MOD] A mobile has its position $x(t) = 3t + 2$ (m, t in s). Where is he to $t = 4$?

Rep. $x(4) = 14$ m.

2. Heart · Method [MOD] A second motive follows $x_2(t) = 5t - 4$. At what moment and where the motives Do they meet?

Rep. $3t + 2 = 5t - 4 \Leftrightarrow t = 3$ s ; position $x = 11$ m.

3. Heart + · Flash [GEO] A velocity vector has coordinates $\begin{pmatrix}3\\4\end{pmatrix}$ (m/s) What is its norm?

Rep. $\sqrt{3^2 + 4^2} = 5$ m/s.

4. Back · Flash [CAL] Calculate $\dfrac{1}{2} - \dfrac{3}{5}$.

Rep. $\dfrac{5}{10} - \dfrac{6}{10} = -\dfrac{1}{10}$.

5. Red thread · Write [DEM] ABC is isocele in A and H is the foot of the height coming from A. What can we do? affirming that H ?

Rep. H is the midpoint of $[BC]$ (The height from the top is also the mediator of the base).

Form 60 game. 8 Apr 2027 Communicating a demonstration: involvement, mutual, equivalence

1. Warming · Flash [DEM] To state the converse of: "if a quadrilateral is a diamond, then its diagonals are perpendicular".

Rep. "If a quadrilateral has its perpendicular diagonals, then it is a diamond."

2. Heart · True/false [DEM] True or false? Justify. The previous converse is true.

Rep. False. Counter-example: a kite (non-loose) has perpendicular diagonals.